FRANKLIN (ZHENYANG) FU

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Education

University of Washington Sep 2024 - Dec 2027

B.S., Electrical and Computer Engineering Seattle

- **Coursework:** Computer Architecture, Embedded Systems, Systems Programming (C/C++), Digital Circuits, Statics, Mechanics of Materials, Kinematics, Probability & Statistics, Data-Driven Modeling and Machine Learning, Foundations of Machine Learning

Publications

- W. Xing*, F. Fu*, M. Han. Knowing the Answer Is Not Enough: Answer-Set Validation in Medical Question Answering. *ACM AI Letters (AILET) - under review. (*equal contribution)*
- A. Wadhwani*, J. Shao*, R. Goenka*, Z. Fu, X. Shen, Q. Xue. Demo of NailSense: Unobtrusive Wireless Smart Nail Stickers for Whole-Hand Posture Tracking. *UIST 2026 Demo Track - conditionally accepted. (*equal contribution)*

Technical Skills

- **Programming**: C, C++, Python (NumPy, Pandas), SystemVerilog
- **Computer Architecture & Digital Design**: pipelined CPU design (forwarding, hazard detection), memory hierarchy & cache analysis, RTL simulation & verification
- **Embedded Systems**: STM32 & ESP32 (ARM Cortex-M), PX4 / Pixhawk, FreeRTOS, Sensor Integration, Motor Control, RTOS Architecture, Device Drivers, Communication Protocols (SPI, I2C, UART, USB)
- **Machine Learning & Vision**: PyTorch, scikit-learn, OpenCV, YOLOv8, LLM evaluation & benchmarking
- **Hardware**: PCB & flex-PCB prototyping, fine-pitch soldering
- **Mechanics & Fabrication**: SolidWorks (CSWA Certified), Fusion 360, statics & stress analysis, mechanism design, 3D printing
- **Development Tools**: Git, Linux
- **Languages:** English (professional proficiency), Chinese (native)

EXPERIENCE

UW Husky Robotics, Drone Subsystem Mar 2026 - Present

Embedded Firmware Developer

- Integrate the PX4 autopilot on a Pixhawk 6X flight controller paired with an NVIDIA Jetson companion computer for onboard compute (in progress), after determining that a standalone STM32 flight controller could not meet the team's computing requirements.
- Brought up a BMI270 6-axis IMU streaming calibrated data at 100 Hz, porting Bosch's SensorAPI through a custom HAL layer and adapting the communication interface from SPI to I2C after tracing a board-level hardware conflict.
- Designed a dual-board prototype (STM32G431 ESC + STM32H753 flight controller) linked over FDCAN, laying the groundwork for field-oriented motor control.

University of Washington Ubicomp Lab Jun 2026 - Aug 2026

Research Assistant

- Built and assembled hardware for NailSense, a flex-PCB wireless smart nail system for whole-hand posture tracking: fine-pitch soldering of nail-scale assemblies and battery charging and handling for the wireless prototypes.
- Owned the fingernail attachment mechanism across three design iterations, diagnosing each failure: silicone substrates rejected adhesives due to low surface energy; adhesive on the electronics face delaminated the flex PCB under peel force during removal; relocating the adhesive to the back surface eliminated PCB stress while improving conformity to the nail.
- Co-authored the resulting UIST 2026 demo submission (conditionally accepted).

Zhejiang University, Binjiang Institute Jan 2026 - Mar 2026

Research Assistant

- Designed Diabetes-FCT, a 1,020-item behavioral benchmark separating answer selection from answer-set validation, transforming 255 clinical seeds into paired factual-control, no-valid-option, and invalid-meta-option tasks, plus an

independent 255-item fictional-premise task.
- Built an automated zero-shot evaluation pipeline in Python using regex parsing and local model deployment via Ollama, stress-testing five 7–8B models.
- Surfaced a factual-accuracy versus answer-set-validity gap: Llama-3.1-8B scored 88.2% on factual control but 23.5% when no valid option existed, while Mistral-7B fell from 81.6% to 31.0% - evidence that the two capabilities are behaviorally
distinct.

UW Craft² Student Research Lab Sep 2024 - May 2026

Mechanics Leader
- Designed the hardware for HaptiNav, a haptic navigation system for blind and low-vision pedestrians: a MagSafe-compatible iPhone attachment with an ESP32-S3 BLE MCU and three directional vibration motors, running 5–7 hours on a 3.7 V Li-Po.
- Built an iOS app using ARKit LiDAR depth and camera pose reconstruction to classify obstacles (ground, drops, stairs, slopes) into an angular free-space map at up to 5 m range, transmitting avoidance commands over BLE.
- Implemented safety-first arbitration between Google Maps route guidance and the local LiDAR safety layer, overriding navigation when obstacles are detected.
- Earlier designed phone-mounted and wrist-mounted vibrotactile compass prototypes, presented at the UW Undergraduate Research Symposium.

PROJECTS

5-Stage Pipelined LEGv8 Processor 2026
- EE 469 Course Project; SystemVerilog, ModelSim
- Designed and verified a 5-stage pipelined LEGv8 CPU implementing an ARM64 subset, including full forwarding and hazard detection; passed all 12 benchmarks, including recursive Fibonacci.
- Built the datapath incrementally - register file, ALU, single-cycle core, then pipeline - validating each stage against predicted cycle-by-cycle waveform traces before integration.
- Reverse-engineered an undocumented cache's associativity, block size, and capacity through black-box access-latency measurement.

BinSense: Touchless Sensing Waste Bin 2026
- ESP32-S3, C++, FreeRTOS
- Built a touchless waste bin as a dual-core FreeRTOS application coordinating four sensors: PIR hand detection to open a stepper-driven lid, ultrasonic fill-level sensing, photoresistor lid-state detection, and temperature/humidity monitoring to flag spoilage, with a 1602 LCD status display.
- Diagnosed a timing-corruption bug in which concurrent ultrasonic pulseIn() measurement broke DHT11 single-wire bit-bang timing; applied interrupt locking to let both tasks coexist.
- Implemented self-calibration and closed-loop actuation confirmation via photoresistor feedback; documented the codebase with Doxygen.

Automated Laser Marking System Feb 2026 - Mar 2026
- Python, OpenCV, YOLOv8, ESP32
- Architected a computer vision pipeline and ESP32-controlled laser marking gantry to localize Eppendorf tubes for medical laboratory sample-tracking automation.
- Combined YOLOv8n high-confidence anchor detection with OpenCV homography transforms to project a 13 × 15 mm 96-well grid, overcoming low-contrast lighting failures.
- Restructured custom CVAT datasets and applied Mosaic and HSV augmentation to improve dark-object recognition.
- Translated perspective-corrected physical coordinates into G-code and stepper pulse commands streamed to the ESP32 over serial.